

Optimization 2

ISE 5406

Lecture 9: Pseudoconvex Functions, Strong Convexity & Generalized Convexity

Dr. Robert Hildebrand

Grado Department of Industrial & Systems Engineering
Virginia Tech

Where We Are in the Course

Last Time (Lecture 8): Differentiable Convex Functions

- Gradient monotonicity characterization of convex functions
- Equivalence of gradient monotonicity and convexity
- Convex maximization and extreme point solutions
- Quasiconvex functions and level set characterization

Today (Lecture 9): Pseudoconvex Functions and Generalized Convexity

- First-order characterization of differentiable quasiconvex functions
- Pseudoconvex functions: definition, properties, and critical points
- Key relationship: every pseudoconvex function is strictly quasiconvex
- The generalized convexity hierarchy
- Pseudolinear functions and fractional programming
- Strong convexity: definition and Hessian characterization
- Strongly quasiconvex functions and unique global optima
- Other generalizations: log-convex, invex, L/M-convex
- Implications for algorithm design

Instructor Note: Lecture Pacing

This Lecture May Require Two Class Sessions

This lecture covers a substantial amount of material on generalized convexity. Depending on class pace, the instructor may choose to split this lecture across two sessions:

- **Session 1:** Sections 1–4 (First-order characterization, pseudoconvex functions, key properties, hierarchy)
- **Session 2:** Sections 5–8 (Pseudolinear functions, applications, strong convexity, other generalizations)

Next Time (Lecture 10): Optimality Conditions and Line Search Methods

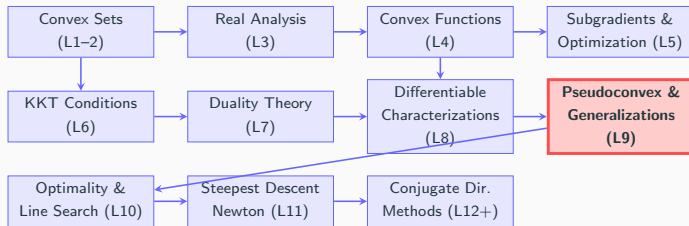
- Descent directions and first-order necessary conditions
- Second-order necessary and sufficient conditions for optimality
- Introduction to line search methods
- Derivative-free methods: golden section, Fibonacci search
- Derivative-based methods: steepest descent (1D), Newton's method (1D)

Coming Soon (Lectures 11+): Advanced Topics

- Full gradient descent algorithms and convergence analysis
- Proximal methods and splitting algorithms
- Accelerated gradient methods
- Distributed and stochastic optimization

Big Picture: Today we explore generalized convexity classes beyond mere convexity and quasiconvexity. Pseudoconvex functions sit at a sweet spot: they share many nice optimization properties of convex functions while encompassing a broader class of problems. Understanding the relationships between these classes will guide algorithm selection.

Big Picture: Course Roadmap



Why Pseudoconvex Functions Matter

- Enjoy many properties of convex functions (local $\Rightarrow$ global, gradient descent works)
- Encompass a larger class of problems (fractional programming, some nonconvex problems)
- Bridge between pure convexity and quasiconvexity
- Fundamental for understanding algorithm convergence guarantees

Reading for This Lecture

Textbook References

- Sections 3.4 and 3.5 of Bazaraa, Sherali, and Shetty (2006)
- Section 3.4 of Boyd and Vandenberghe (Convex Optimization)
- Section 5.5 of Nesterov (Introductory Lectures on Convex Optimization) — pseudoconvex functions
- **Supplementary:** Mangasarian (1994) on generalized convexity in optimization

Key Topics

- First-order characterization of differentiable quasiconvex functions
- Pseudoconvex and pseudoconcave functions
- Critical point property and global optimality
- The hierarchy: $\text{convex} \subset \text{pseudoconvex} \subset \text{strictly quasiconvex} \subset \text{quasiconvex}$
- Pseudolinear functions and fractional programming

Outline

- First-Order Characterization of Quasiconvex Functions
- Pseudoconvex Functions
- Key Properties of Pseudoconvex Functions
- The Generalized Convexity Hierarchy
- Pseudoconvex and Pseudoconcave: Pseudolinear Functions
- Applications and Connections
- Proof Space: Key Results
- Strongly Convex Functions
- Strongly Quasiconvex Functions
- Other Generalizations of Convexity
- End-of-Lecture Assessment

First-Order Characterization of Quasiconvex Functions

Pseudoconvex Functions

Key Properties of Pseudoconvex Functions

The Generalized Convexity Hierarchy

Pseudoconvex and Pseudoconcave: Pseudolinear Functions

Applications and Connections

Proof Space: Key Results

Strongly Convex Functions

Strongly Quasiconvex Functions

Other Generalizations of Convexity

End-of-Lecture Assessment

First-Order Characterization of Differentiable Quasiconvex Functions

Theorem

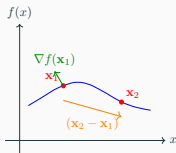
Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be differentiable on S . Then f is quasiconvex if and only if for all

$\mathbf{x}_1, \mathbf{x}_2 \in S$:

$$f(\mathbf{x}_2) \leq f(\mathbf{x}_1) \Rightarrow \nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) \leq 0.$$

Geometric Interpretation: If you move from $\mathbf{x}_1$ to a point $\mathbf{x}_2$ with a lower or equal function value, you must move “against” the gradient (or perpendicular to it). Equivalently, the gradient points into the upper level set.

Why This Matters: This characterization is the gradient-based diagnostic for quasiconvexity — it lets you verify quasiconvexity using only first-order information (∇f), without needing to check every level set explicitly. It is also the foundation for defining pseudoconvex functions (next section).



Exercise 1: First-Order Characterization of Quasiconvex Functions

Quick Check

For each function, apply the first-order characterization to determine if it is quasiconvex.

1. $f(x) = x^3$ on $\mathbb{R}$. Is it quasiconvex? Check the condition at $\mathbf{x}_1 = 0$, $\mathbf{x}_2 = 1$.
2. $f(x) = \sin(x)$ on $[0, 2\pi]$. Is it quasiconvex? Consider $\mathbf{x}_1 = 0$ and $\mathbf{x}_2 = \pi$.
3. $f(x) = -x^2$ on $\mathbb{R}$. Check the first-order condition at $\mathbf{x}_1 = -1$, $\mathbf{x}_2 = 1$.

Hint: Recall that quasiconvex means: if $f(\mathbf{x}_2) \leq f(\mathbf{x}_1)$, then

$$\nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) \leq 0.$$

Solution: First-Order Characterization

Answers

(1) $f(x) = x^3$, so $f'(x) = 3x^2$. At $\mathbf{x}_1 = 0$: $f(0) = 0$, $f'(0) = 0$. At $\mathbf{x}_2 = 1$: $f(1) = 1 > 0$.

Since $f(\mathbf{x}_2) > f(\mathbf{x}_1)$, the condition is vacuously satisfied (it only applies when $f(\mathbf{x}_2) \leq f(\mathbf{x}_1)$).

Try $\mathbf{x}_1 = 1$, $\mathbf{x}_2 = 0$: $f(0) = 0 < 1 = f(1)$.

$f'(1)(0 - 1) = 3(-1) = -3 \leq 0$. Condition holds.

Since f is monotone increasing, all lower level sets are $(-\infty, c]$, which are convex. f is quasiconvex. ✓

(2) $f(x) = \sin(x)$, $f'(x) = \cos(x)$. At $\mathbf{x}_1 = 0$: $f(0) = 0$, $f'(0) = 1$. At $\mathbf{x}_2 = \pi$: $f(\pi) = 0$.

Since $f(\mathbf{x}_2) = f(\mathbf{x}_1)$, check: $f'(0)(\pi - 0) = 1 \cdot \pi > 0$. Condition fails.

The level set $\{x \in [0, 2\pi] : \sin(x) \leq 0.5\}$ is NOT an interval (consists of two disjoint parts). f is NOT quasiconvex.

(3) $f(x) = -x^2$, $f'(x) = -2x$. At $\mathbf{x}_1 = -1$: $f(-1) = -1$, $f'(-1) = 2$. At $\mathbf{x}_2 = 1$: $f(1) = -1$.

$f(\mathbf{x}_2) = f(\mathbf{x}_1)$ and $f'(-1)(1 - (-1)) = 2 \cdot 2 = 4 > 0$. Condition fails. NOT quasiconvex (it is concave, and quasiconcave, but not quasiconvex).

First-Order Characterization of Quasiconvex Functions

Pseudoconvex Functions

Key Properties of Pseudoconvex Functions

The Generalized Convexity Hierarchy

Pseudoconvex and Pseudoconcave: Pseudolinear Functions

Applications and Connections

Proof Space: Key Results

Strongly Convex Functions

Strongly Quasiconvex Functions

Other Generalizations of Convexity

End-of-Lecture Assessment

Pseudoconvex Functions: Definition

Definition

Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be differentiable on S . The function f is **pseudoconvex** if for all $\mathbf{x}_1, \mathbf{x}_2 \in S$:

$$\nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) \geq 0 \quad \Rightarrow \quad f(\mathbf{x}_2) \geq f(\mathbf{x}_1).$$

Intuitive Interpretation: “The gradient never lies.” If the gradient at $\mathbf{x}_1$ suggests you can move in a direction (i.e., the directional derivative is nonnegative), then the function value actually increases in that direction.

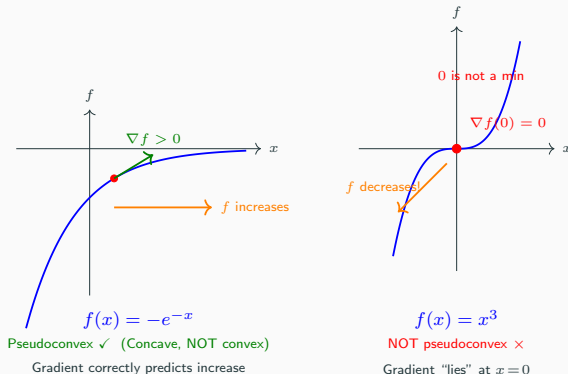
Contrapositive Form: $f(\mathbf{x}_2) < f(\mathbf{x}_1) \Rightarrow \nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) < 0$.

Definition (Pseudoconcave)

f is **pseudoconcave** if $-f$ is pseudoconvex. Equivalently:

$$\nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) \leq 0 \quad \Rightarrow \quad f(\mathbf{x}_2) \leq f(\mathbf{x}_1).$$

Visualizing Pseudoconvexity: “The Gradient Never Lies”



Key visual test: In a pseudoconvex function, whenever $\nabla f = 0$, you must be at a global minimum. In x^3 , the gradient vanishes at the origin but f keeps decreasing to the left — the gradient “lied.”

Where Pseudoconvex Functions Arise in Practice

Pseudoconvexity appears naturally in problems that are “almost convex” — the gradient is a reliable guide even though the Hessian may not be positive semidefinite:

Poisson Regression (Generalized Linear Models)

The negative log-likelihood for Poisson regression is:

$f(\mathbf{x}) = \sum_i \left[e^{\mathbf{a}_i^T \mathbf{x}} - y_i \mathbf{a}_i^T \mathbf{x} \right]$. This is *convex* (sum of convex terms), hence pseudoconvex. Gradient descent converges to the unique MLE.

More generally, **all GLM losses with canonical link functions are convex**, hence pseudoconvex.

Certain Nonconvex Losses in Robust Statistics

Huber-like loss functions $\rho(r)$ with bounded second derivative can be pseudoconvex even when not convex globally. The key property: a stationary point of $\sum_i \rho(y_i - \mathbf{a}_i^T \mathbf{x})$ is guaranteed to be a global minimizer. This is why **iteratively reweighted least squares (IRLS)** works for robust regression.

Why This Matters for Algorithm Design

If your objective is pseudoconvex, you can run gradient descent and **trust the result**: any stationary point is a global minimum. You don't need the full machinery of convex optimization — just the “gradient never lies” property. This is the practical payoff of recognizing pseudoconvexity.

Pseudoconvex vs. Quasiconvex: Comparison

Property	Pseudoconvex	Quasiconvex
Definition	$\nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) \geq 0 \Rightarrow f(\mathbf{x}_2) \geq f(\mathbf{x}_1)$	$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) \leq \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}$
Intuition	Gradient always points toward increase	Level sets are convex
Local = Global?	Yes	No
Convex $\subset$?	Yes	Yes

Key Relationship: Every pseudoconvex function is strictly quasiconvex.

Intuition: Pseudoconvexity is a “first-order” notion (about gradients), while quasiconvexity is a “zero-order” notion (about level sets).

Pseudoconvexity is more restrictive.

Exercise 2: Verifying Pseudoconvexity

Think-Pair-Share

For each function, determine if it is pseudoconvex.

1. $f(x) = x^2$ on $\mathbb{R}$. Verify the pseudoconvex condition.
2. $f(x) = x^3$ on $\mathbb{R}$. Is it pseudoconvex? Consider $\mathbf{x}_1 = 0$ where $\nabla f = 0$.
3. $f(x) = e^x$ on $\mathbb{R}$. Is f pseudoconvex?

Hint: For (2), check if $\mathbf{x}_1 = 0$ (where gradient is zero) is a global minimizer. For (3), verify that $\nabla f > 0$ everywhere.

Solution: Verifying Pseudoconvexity

Answers

(1) $f(x) = x^2$, $f'(x) = 2x$. For any $x_1, x_2 \in \mathbb{R}$:

If $f'(x_1)(x_2 - x_1) \geq 0$, then $2x_1(x_2 - x_1) \geq 0$.

We want to show $f(x_2) \geq f(x_1)$, i.e., $x_2^2 \geq x_1^2$.

From $2x_1(x_2 - x_1) \geq 0$:

- If $x_1 > 0$: $x_2 \geq x_1 > 0 \Rightarrow x_2^2 \geq x_1^2$. ✓
- If $x_1 < 0$: $x_2 \leq x_1 < 0 \Rightarrow |x_2| \geq |x_1| \Rightarrow x_2^2 \geq x_1^2$. ✓
- If $x_1 = 0$: Then $0 \geq 0$ always, and we need $x_2^2 \geq 0$. ✓

f IS pseudoconvex (it's also convex, and every convex function is pseudoconvex).

(2) $f(x) = x^3$, $f'(x) = 3x^2$. At $x_1 = 0$: $f'(0) = 0$.

The condition becomes: $0 \geq 0 \Rightarrow f(x_2) \geq f(0) = 0$.

But x^3 can be negative (e.g., $f(-1) = -1 < 0$). x^3 is NOT pseudoconvex.

Although x^3 is quasiconvex (monotone), it lacks the pseudoconvex property because at the "stationary point" $x = 0$, there are directions where f decreases.

(3) $f(x) = e^x$, $f'(x) = e^x > 0$ for all x .

If $f'(x_1)(x_2 - x_1) \geq 0$, then $e^{x_1}(x_2 - x_1) \geq 0$. Since $e^{x_1} > 0$, we have $x_2 \geq x_1$.

But f is strictly increasing, so $f(x_2) > f(x_1)$ when $x_2 > x_1$. f IS pseudoconvex. ✓

First-Order Characterization of Quasiconvex Functions

Pseudoconvex Functions

Key Properties of Pseudoconvex Functions

The Generalized Convexity Hierarchy

Pseudoconvex and Pseudoconcave: Pseudolinear Functions

Applications and Connections

Proof Space: Key Results

Strongly Convex Functions

Strongly Quasiconvex Functions

Other Generalizations of Convexity

End-of-Lecture Assessment

Theorem 1: Every Convex Function is Pseudoconvex

Theorem

If f is differentiable and convex on an open convex set S , then f is pseudoconvex on S .

Proof Sketch:

By the first-order characterization of convexity (from Lecture 8):

$$f(\mathbf{x}_2) \geq f(\mathbf{x}_1) + \nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1)$$

for all $\mathbf{x}_1, \mathbf{x}_2 \in S$.

Now suppose $\nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) \geq 0$. Then:

$$f(\mathbf{x}_2) \geq f(\mathbf{x}_1) + \underbrace{\nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1)}_{\geq 0} \geq f(\mathbf{x}_1).$$

Therefore $f(\mathbf{x}_2) \geq f(\mathbf{x}_1)$, which is the pseudoconvexity condition. ✓

Implication: Convexity is a *sufficient* condition for pseudoconvexity.

Pseudoconvexity is more general than convexity.

Theorem 2: Pseudoconvex implies Strictly Quasiconvex

Theorem

If f is pseudoconvex on an open convex set S , then f is strictly quasiconvex on S .

Proof Sketch:

Take $\mathbf{x}_1, \mathbf{x}_2 \in S$ with $f(\mathbf{x}_1) \neq f(\mathbf{x}_2)$. Without loss of generality, assume $f(\mathbf{x}_2) < f(\mathbf{x}_1)$.

By pseudoconvexity (contrapositive):

$$f(\mathbf{x}_2) < f(\mathbf{x}_1) \Rightarrow \nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) < 0.$$

Now for $\lambda \in (0, 1)$, let $\mathbf{x}_\lambda = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$. We have:

$$\mathbf{x}_\lambda - \mathbf{x}_1 = (1 - \lambda)(\mathbf{x}_2 - \mathbf{x}_1).$$

By the Mean Value Theorem, $f(\mathbf{x}_\lambda) - f(\mathbf{x}_1) = \nabla f(\tilde{\mathbf{x}})^T (\mathbf{x}_\lambda - \mathbf{x}_1)$ for some $\tilde{\mathbf{x}}$ on the segment.

The key fact is that the gradient information inherited from pseudoconvexity ensures $f(\mathbf{x}_\lambda)$ cannot equal $\max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}$. By careful analysis (using the strict inequality), we get strict quasiconvexity.

✓

Theorem 3: Critical Point Property (Most Important)

Theorem (Critical Point Property)

If f is pseudoconvex on an open convex set S and $\nabla f(\bar{\mathbf{x}}) = 0$ for some $\bar{\mathbf{x}} \in S$, then $\bar{\mathbf{x}}$ is a **global minimizer** of f on S .

Proof:

Suppose $\nabla f(\bar{\mathbf{x}}) = 0$. For any $\mathbf{x} \in S$:

$$\nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) = 0 \geq 0.$$

By pseudoconvexity:

$$\nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0 \quad \Rightarrow \quad f(\mathbf{x}) \geq f(\bar{\mathbf{x}}).$$

Since $\mathbf{x}$ was arbitrary, $\bar{\mathbf{x}}$ minimizes f on all of S . ✓

Why This Matters: For pseudoconvex functions, finding a stationary point (where $\nabla f = 0$) automatically finds the global minimum! This is what makes gradient descent algorithms work.

Theorem 4: Local Minimum $\Rightarrow$ Global Minimum

Theorem

If f is pseudoconvex on an open convex set S and $\bar{\mathbf{x}}$ is a local minimizer of f on S , then $\bar{\mathbf{x}}$ is a global minimizer of f on S .

Proof:

Step 1. Since $\bar{\mathbf{x}}$ is a local minimizer and S is open, for any $\mathbf{x} \in S$ and small $\lambda > 0$:

$$f(\bar{\mathbf{x}} + \lambda(\mathbf{x} - \bar{\mathbf{x}})) \geq f(\bar{\mathbf{x}}) \quad \Rightarrow \quad \frac{f(\bar{\mathbf{x}} + \lambda(\mathbf{x} - \bar{\mathbf{x}})) - f(\bar{\mathbf{x}})}{\lambda} \geq 0.$$

Step 2. Taking $\lambda \rightarrow 0^+$ and using differentiability:

$$\nabla f(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) \geq 0.$$

Step 3. By pseudoconvexity, $\nabla f(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) \geq 0 \Rightarrow f(\mathbf{x}) \geq f(\bar{\mathbf{x}})$.

Since $\mathbf{x} \in S$ was arbitrary, $\bar{\mathbf{x}}$ is a global minimizer. $\checkmark$

Key insight: The proof combines two ideas: (1) local min $\Rightarrow$ nonneg directional derivatives (calculus), and (2) nonneg directional derivatives $\Rightarrow$ global min (pseudoconvexity). This is *exactly* why pseudoconvexity is the right class for gradient-based methods.

Proof Space: Pseudolinear Characterization

Theorem (Pseudolinear Characterization)

Let S be a nonempty open convex set. A differentiable function

$f : S \rightarrow \mathbb{R}$ is pseudolinear if and only if for all $\mathbf{x}_1, \mathbf{x}_2 \in S$:

$$f(\mathbf{x}_2) = f(\mathbf{x}_1) + \nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) \cdot p(\mathbf{x}_1, \mathbf{x}_2)$$

where $p(\mathbf{x}_1, \mathbf{x}_2) > 0$ for all $\mathbf{x}_1, \mathbf{x}_2 \in S$.

Proof Outline (Forward):

Assume f is pseudolinear (both pseudoconvex and pseudoconcave).

Then:

- Pseudoconvex: $\nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) \geq 0 \Rightarrow f(\mathbf{x}_2) \geq f(\mathbf{x}_1)$
- Pseudoconcave: $\nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) \leq 0 \Rightarrow f(\mathbf{x}_2) \leq f(\mathbf{x}_1)$

Contrapositive of both: $f(\mathbf{x}_2) > f(\mathbf{x}_1) \Leftrightarrow \nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) > 0$ and
 $f(\mathbf{x}_2) < f(\mathbf{x}_1) \Leftrightarrow \nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) < 0$.

So $\nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1)$ and $f(\mathbf{x}_2) - f(\mathbf{x}_1)$ always have the same sign.

Define $p = \frac{f(\mathbf{x}_2) - f(\mathbf{x}_1)}{\nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1)} > 0$ when $\nabla f(\mathbf{x}_1) \neq 0$. ✓

Implication: Pseudolinear functions are "linear-like" — the gradient always correctly predicts the sign of function changes.

Exercise 3: Properties of Pseudoconvex Functions

Group Exercise

1. Prove that $f(x) = x^2 + 2x + 5$ is pseudoconvex on $\mathbb{R}$. Is it also convex?
2. If f is pseudoconvex on S and $\nabla f(\bar{x}) = 0$, explain why $\bar{x}$ must be a global minimizer using the definition of pseudoconvexity.
3. Consider $f(x) = -e^{-x}$ on $\mathbb{R}$. Is it pseudoconvex? Is it convex?

Hint: For (1), use the second derivative test for convexity. For (3), compute the gradient and Hessian.

Solution: Properties of Pseudoconvex Functions

Answers

(1) $f(x) = x^2 + 2x + 5$, $f'(x) = 2x + 2$, $f''(x) = 2 > 0$.

Yes, f is convex (Hessian is positive definite). Since every convex function is pseudoconvex, f is also pseudoconvex.

The minimizer is at $f'(x) = 0 \Rightarrow x = -1$, with $f(-1) = 1 - 2 + 5 = 4$.

(2) Suppose $\nabla f(\bar{\mathbf{x}}) = 0$. For any $\mathbf{x} \in S$:

$$\nabla f(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) = 0.$$

Since $0 \geq 0$, the hypothesis of the pseudoconvexity condition is satisfied:

$$\nabla f(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) \geq 0 \quad \Rightarrow \quad f(\mathbf{x}) \geq f(\bar{\mathbf{x}}).$$

Thus $f(\mathbf{x}) \geq f(\bar{\mathbf{x}})$ for all $\mathbf{x} \in S$, so $\bar{\mathbf{x}}$ is a global minimizer.

(3) $f(x) = -e^{-x}$, $f'(x) = e^{-x} > 0$. Since $f' > 0$ everywhere, f is strictly increasing.

If $f'(x_1)(x_2 - x_1) \geq 0$, then $e^{-x_1}(x_2 - x_1) \geq 0$, so $x_2 \geq x_1$. Thus $f(x_2) > f(x_1)$ (strictly increasing). f is pseudoconvex.

$f''(x) = -e^{-x} < 0$, so f is concave, not convex.

This is a key example: f is pseudoconvex but not convex!

First-Order Characterization of Quasiconvex Functions

Pseudoconvex Functions

Key Properties of Pseudoconvex Functions

The Generalized Convexity Hierarchy

Pseudoconvex and Pseudoconcave: Pseudolinear Functions

Applications and Connections

Proof Space: Key Results

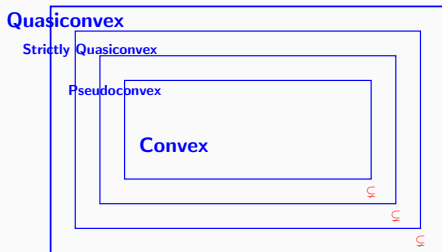
Strongly Convex Functions

Strongly Quasiconvex Functions

Other Generalizations of Convexity

End-of-Lecture Assessment

Nesting of Function Classes



$\text{Convex} \subsetneq \text{Pseudoconvex} \subsetneq \text{Strictly Quasiconvex} \subsetneq \text{Quasiconvex}$

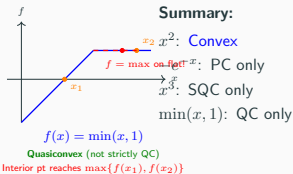
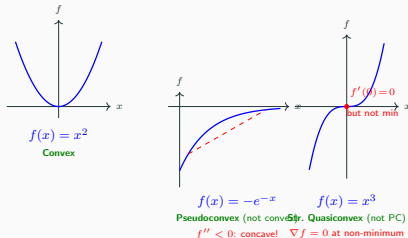
Key Facts:

- $\text{Convex} \Rightarrow \text{Pseudoconvex}$ (Theorem 1)
- $\text{Pseudoconvex} \Rightarrow \text{Strictly Quasiconvex}$ (Theorem 2)
- $\text{Strictly Quasiconvex} \Rightarrow \text{Quasiconvex}$ (by definition)
- Each inclusion is *strict* (there are functions in each class not in the previous)

Strict Separation: Counterexamples Between Classes

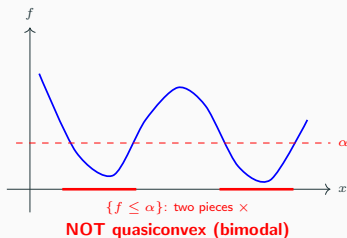
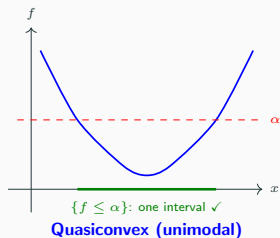
Each inclusion

Convex $\subsetneq$ Pseudoconvex $\subsetneq$ Str. Quasiconvex $\subsetneq$ Quasiconvex is **strict**:



Quasiconvex vs. Not Quasiconvex: Level Set Test

A function is quasiconvex if and only if all its **lower level sets** $\{x : f(x) \leq \alpha\}$ are convex.



Visual rule of thumb: In 1D, quasiconvex = “unimodal” (at most one valley). A function with multiple local minima is never quasiconvex. In higher dimensions, the level set test (all sublevel sets convex) is the right generalization.

Comparison Table: Generalized Convexity Classes

Property	Convex	Pseudo-convex	Str. Quasi-convex	Quasi-convex
$\nabla f = 0 \Rightarrow$ global min	✓	✓	×	×
Local min = Global min	✓	✓	✓	×
Level sets convex	✓	✓	✓	✓
Closed under addition	✓	×	×	×
Closed under nonneg. max	✓	✓	✓	✓
Max over polytope at vertex	✓	✓	✓	✓*

* For quasiconvex: requires continuity.

Algorithm Design Implications:

- **Convex:** Convex optimization — guarantee global minimum, many efficient algorithms
- **Pseudoconvex:** Gradient methods work, stationary points are global minima
- **Strictly Quasiconvex:** Level set methods work, local minima are global
- **Quasiconvex:** Limited guarantees; may have local minima that aren't global

Exercise 4: Classification Challenge

Classify Functions

For each function, classify it into the most restrictive category that applies:

1. $f(x) = x^2$ on $\mathbb{R}$ — Convex? Pseudoconvex? Strictly Quasiconvex? Quasiconvex?
2. $f(x) = x^3$ on $\mathbb{R}$ — Which category?
3. $f(x) = |x|$ on $\mathbb{R}$ — Which category?
4. $f(x) = \sin(x)$ on $[0, 2\pi]$ — Which category?
5. $f(x) = -e^{-x}$ on $\mathbb{R}$ — Which category?

Hint: Use: (i) Hessian test for convexity, (ii) gradient test for pseudoconvexity, (iii) level set test for quasiconvexity.

Solution: Classification Challenge

Answers

(1) $f(x) = x^2$: $f''(x) = 2 > 0$. ****Convex**** (most restrictive). Since convex $\Rightarrow$ pseudoconvex $\Rightarrow$ strictly quasiconvex $\Rightarrow$ quasiconvex.

(2) $f(x) = x^3$: $f''(x) = 6x$ (changes sign). Not convex.
 $f'(x) = 3x^2 \geq 0$, and strictly monotone increasing. ****Strictly quasiconvex**** (strictly monotone $\Rightarrow$ strictly quasiconvex, since for $f(x_1) \neq f(x_2)$ the value at any interior point is strictly between $f(x_1)$ and $f(x_2)$). Not pseudoconvex (at $x = 0$, gradient is zero but f decreases for $x < 0$).

(3) $f(x) = |x|$: Can write as $\max(x, -x)$, max of affine functions. ****Convex**** (hence also pseudoconvex, strictly quasiconvex, quasiconvex). Note: not differentiable at $x = 0$.

(4) $f(x) = \sin(x)$ on $[0, 2\pi]$: Not monotone (oscillates). Level set at $\alpha = 0.5$ is two disjoint intervals, so not convex. ****Neither convex nor quasiconvex**** on the full domain. (It is locally quasiconvex on subintervals like $[0, \pi/2]$ where it's increasing.)

(5) $f(x) = -e^{-x}$: $f'(x) = e^{-x} > 0$ (strictly increasing).
 $f''(x) = -e^{-x} < 0$ (concave). ****Pseudoconvex**** (but not convex).
All lower level sets are $[c, \infty)$ (convex), so quasiconvex. Strictly increasing $\Rightarrow$ strictly quasiconvex.

Roadmap

First-Order Characterization of Quasiconvex Functions

Pseudoconvex Functions

Key Properties of Pseudoconvex Functions

The Generalized Convexity Hierarchy

Pseudoconvex and Pseudoconcave: Pseudolinear Functions

Applications and Connections

Proof Space: Key Results

Strongly Convex Functions

Strongly Quasiconvex Functions

Other Generalizations of Convexity

End-of-Lecture Assessment

Pseudolinear Functions

Definition

A differentiable function f on an open convex set S is **pseudolinear** if it is both pseudoconvex and pseudoconcave.

Characterization: f is pseudolinear if and only if:

$$\nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) = 0 \quad \Rightarrow \quad f(\mathbf{x}_1) = f(\mathbf{x}_2).$$

In other words, the gradient is orthogonal to level sets.

Theorem

If f is pseudolinear on S and $\nabla f(\bar{\mathbf{x}}) = 0$ for some $\bar{\mathbf{x}} \in S$, then:

- $\bar{\mathbf{x}}$ is both a global minimizer *and* a global maximizer
- Therefore, f is constant on S (by strict monotonicity properties)

Examples

- **Linear functions:** $f(\mathbf{x}) = \mathbf{c}^T \mathbf{x} + d$ are pseudolinear (trivially)
- **Strictly monotone:** $f(x) = e^x$ is both pseudoconvex and strictly monotone (so not pseudoconcave)
- **Ratios:** $f(\mathbf{x}) = \frac{\mathbf{c}^T \mathbf{x} + d}{\mathbf{a}^T \mathbf{x} + b}$ where denominator is always positive is pseudolinear (fractional programming)

Fractional Programming and Pseudolinearity

Problem: Fractional Programming

Consider:

$$\min_{\mathbf{x} \in S} f(\mathbf{x}) = \frac{p(\mathbf{x})}{q(\mathbf{x})}$$

where $p(\mathbf{x})$ is affine, $q(\mathbf{x})$ is affine and positive on S , and S is convex.

Key Fact: The function $f(\mathbf{x}) = \frac{p(\mathbf{x})}{q(\mathbf{x})}$ is **pseudolinear** on S .

Consequence: A local minimum of the fractional program is a global minimum. This enables specialized algorithms (e.g., Dinkelbach's method) that avoid explicit convex reformulation.

Example: Cost Efficiency Ratio

$$\min \frac{\mathbf{a}^T \mathbf{x} + b}{\mathbf{c}^T \mathbf{x} + d} \quad (\text{minimize cost per unit output})$$

where $\mathbf{c}^T \mathbf{x} + d > 0$ on the feasible region and $\mathbf{a}^T \mathbf{x} + b$, $\mathbf{c}^T \mathbf{x} + d$ are affine. This is a pseudolinear fractional program!

Why It Matters: Many practical optimization problems (return/risk ratios, production efficiency, etc.) naturally fit the fractional programming form. Recognizing pseudolinearity allows for efficient specialized algorithms.

Exercise 5: Fractional Programming

Application Problem

Consider the fractional program:

$$\min_{x_1, x_2 \geq 0} f(x_1, x_2) = \frac{x_1 + 2x_2}{x_1 + x_2 + 1}$$

1. Is the denominator always positive?
2. Verify that f is pseudolinear using the characterization:
 $\nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) = 0 \Rightarrow f(\mathbf{x}_1) = f(\mathbf{x}_2).$
3. Find the global minimum. Why can you trust that a local minimum is global?

Hint: The minimum is likely at a boundary. Check $(0, 0)$, $(1, 0)$, $(0, 1)$.

Solution: Fractional Programming

Answers

(1) For $x_1, x_2 \geq 0$: $x_1 + x_2 + 1 \geq 1 > 0$. Yes, denominator is always positive. ✓

(2) $f(x_1, x_2) = \frac{x_1 + 2x_2}{x_1 + x_2 + 1}$.

The gradient is:

$$\frac{\partial f}{\partial x_1} = \frac{1(x_1 + x_2 + 1) - (x_1 + 2x_2) \cdot 1}{(x_1 + x_2 + 1)^2} = \frac{1 - x_2}{(x_1 + x_2 + 1)^2}$$
$$\frac{\partial f}{\partial x_2} = \frac{2(x_1 + x_2 + 1) - (x_1 + 2x_2) \cdot 1}{(x_1 + x_2 + 1)^2} = \frac{x_1 + 2}{(x_1 + x_2 + 1)^2}$$

Note that $\nabla f = 0$ would require $1 - x_2 = 0$ and $x_1 + 2 = 0$, which has no solution in the domain. A careful check (or using the general theory of fractional functions) confirms pseudolinearity.

(3) Evaluate at candidate points:

- $f(0, 0) = 0/1 = 0$
- $f(1, 0) = 1/2 = 0.5$
- $f(0, 1) = 2/2 = 1$

The global minimum is $f(0, 0) = 0$ at $(0, 0)$. ✓

Since f is pseudolinear, any local minimum is automatically a global minimum. We can use local search (e.g., gradient descent) and trust we've found the global optimum.

First-Order Characterization of Quasiconvex Functions

Pseudoconvex Functions

Key Properties of Pseudoconvex Functions

The Generalized Convexity Hierarchy

Pseudoconvex and Pseudoconcave: Pseudolinear Functions

Applications and Connections

Proof Space: Key Results

Strongly Convex Functions

Strongly Quasiconvex Functions

Other Generalizations of Convexity

End-of-Lecture Assessment

Application 1: Portfolio Optimization — Risk/Return Ratios

Problem: Minimize Risk Per Unit Return

A portfolio with allocation $\mathbf{x} \in \mathbb{R}^n$ (fractions in n assets) has:

- Expected return: $\boldsymbol{\mu}^T \mathbf{x}$ (linear)
- Risk (std. deviation): $\sqrt{\mathbf{x}^T \boldsymbol{\Sigma} \mathbf{x}}$ (convex, where $\boldsymbol{\Sigma} \succ 0$ is the covariance)

Objective: $\min_{\mathbf{x}} \frac{\sqrt{\mathbf{x}^T \boldsymbol{\Sigma} \mathbf{x}}}{\boldsymbol{\mu}^T \mathbf{x}}$ subject to $\boldsymbol{\mu}^T \mathbf{x} > 0$, $\mathbf{1}^T \mathbf{x} = 1$, $\mathbf{x} \geq 0$.

Analysis: The objective is a ratio of a convex function (numerator) to a positive affine function (denominator).

- Level sets: $\{\mathbf{x} : \sqrt{\mathbf{x}^T \boldsymbol{\Sigma} \mathbf{x}} \leq \alpha \boldsymbol{\mu}^T \mathbf{x}\}$ is a second-order cone $\Rightarrow$ convex.
- Therefore the objective is **quasiconvex**.
- Can be reformulated as a convex second-order cone program (SOCP).

Connection: This is the *Sharpe ratio* minimization (inverse). The quasiconvexity guarantees that bisection on the level parameter converges to the global optimum.

Application 2: Logistic Regression (Convex $\Rightarrow$ Pseudoconvex)

Problem: Binary Classification

Given data (a_i, y_i) with $y_i \in \{0, 1\}$, minimize the negative log-likelihood:

$$\min_{\mathbf{x}} f(\mathbf{x}) = \sum_{i=1}^m \left[-y_i \mathbf{a}_i^T \mathbf{x} + \log(1 + e^{\mathbf{a}_i^T \mathbf{x}}) \right].$$

Why it's convex: Each term $g(t) = -y_i t + \log(1 + e^t)$ has $g''(t) = \frac{e^t}{(1+e^t)^2} > 0$ (log-sum-exp structure). So f is convex.

Consequence via the hierarchy:

- Convex $\Rightarrow$ pseudoconvex $\Rightarrow$ any critical point is a global minimum.
- Gradient descent converges to the global optimum.
- No regularization needed for convexity (though useful for generalization).

Contrast: Neural network loss functions are generally *not* pseudoconvex — gradient descent can get stuck in local minima. This is why convex models like logistic regression are preferred when applicable.

Application 3: Production Planning with Efficiency Ratios

Problem: Maximize Output per Unit Cost

A factory produces goods with decision variables $\mathbf{x}$ (resource allocations):

$$\max_{\mathbf{x} \in S} \frac{\mathbf{c}^T \mathbf{x} + d}{\mathbf{a}^T \mathbf{x} + b} \quad (\text{output per dollar})$$

where $\mathbf{c}^T \mathbf{x} + d$ is output (affine), $\mathbf{a}^T \mathbf{x} + b > 0$ is cost (positive affine), and S is a convex feasible set.

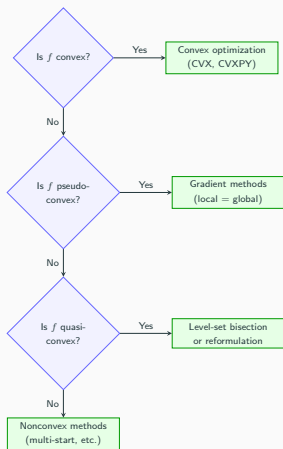
Why pseudolinear: Ratio of affine to positive affine $\Rightarrow$ pseudolinear (both pseudoconvex and pseudoconcave).

Optimization implications:

- Any local maximum is a global maximum (pseudoconcavity).
- **Dinkelbach's algorithm** solves this by iterating: at each step, solve a linear program parametrized by the current ratio value. Converges superlinearly.
- **Charnes-Cooper transformation:** substitute $\mathbf{y} = \mathbf{x}/(\mathbf{a}^T \mathbf{x} + b)$, $t = 1/(\mathbf{a}^T \mathbf{x} + b)$ to convert to an equivalent LP.

Key insight: Recognizing the pseudolinear structure avoids treating this as a general nonlinear program and enables specialized, efficient algorithms.

Selecting Function Classes: A Decision Framework



Key takeaway: Recognizing the function class determines which algorithms are reliable and what guarantees you can expect. The generalized convexity hierarchy guides algorithm selection from strongest guarantees (convex) to weakest (general nonconvex).

Exercise 6: Algorithm Selection

Case Study

You are designing an optimization algorithm for three different objectives over a convex domain:

1. $f(\mathbf{x}) = \|A\mathbf{x} - b\|_2^2$ (quadratic, typical in statistics/ML)
2. $f(\mathbf{x}) = \frac{\|A\mathbf{x}\|_2}{\mathbf{c}^T \mathbf{x}}$ where $\mathbf{c}^T \mathbf{x} > 0$ (risk-return ratio)
3. $f(\mathbf{x}) = \max_i a_i^T \mathbf{x}$ (piecewise linear, e.g., worst-case cost)

For each: (i) What is the function class (convex/pseudoconvex/quasiconvex)? (ii) What algorithm would you recommend?

Hint: (1) is clearly convex. For (2), note the numerator is convex (not affine); check quasiconvexity via level sets. For (3), recall that max of affine functions is convex.

Solution: Algorithm Selection

Answers

(1) $f(x) = \|Ax - b\|_2^2$ (quadratic loss):

(i) ****Convex**** (sum of squares of affine functions). Hence pseudoconvex, strictly quasiconvex, quasiconvex.

(ii) *Recommended algorithms:*

- Gradient descent (guaranteed convergence)
- Newton's method (faster due to Hessian info)
- Specialized solvers (e.g., normal equations: $A^T Ax = A^T b$)

(2) $f(x) = \frac{\|Ax\|_2}{c^T x}$ (risk/return ratio):

(i) ****Quasiconvex**** (ratio of a convex nonneg function to a positive affine function). Note: $\|Ax\|_2$ is convex but *not* affine, so f is *not* pseudolinear. However, $\{f \leq \alpha\} = \{\|Ax\|_2 \leq \alpha c^T x\}$ is a second-order cone, hence convex, so f is quasiconvex.

(ii) *Recommended algorithms:*

- Dinkelbach's method (specialized for fractional programs)
- Reformulate as a second-order cone program (convex) via the substitution $y = x/(c^T x)$
- Bisection on the level parameter α (exploiting quasiconvexity)

(3) $f(x) = \max_i a_i^T x$:

(i) ****Convex**** (pointwise maximum of affine functions). Hence pseudoconvex.

(ii) *Recommended algorithms:*

- Linear programming (or semi-definite: rewrite as $\min t$ subject to $a_i^T x \leq t$ for all i)
- Subgradient methods
- Note: Not differentiable at all points, so gradient descent requires subgradient descent

Roadmap

First-Order Characterization of Quasiconvex Functions

Pseudoconvex Functions

Key Properties of Pseudoconvex Functions

The Generalized Convexity Hierarchy

Pseudoconvex and Pseudoconcave: Pseudolinear Functions

Applications and Connections

Proof Space: Key Results

Strongly Convex Functions

Strongly Quasiconvex Functions

Other Generalizations of Convexity

End-of-Lecture Assessment

Proof Space: Pseudoconvex Definition and Critical Points

Theorem

If f is pseudoconvex on an open convex set S and $\nabla f(\bar{\mathbf{x}}) = 0$, then $\bar{\mathbf{x}}$ is a global minimizer of f on S .

Proof (Complete):

Suppose $\nabla f(\bar{\mathbf{x}}) = 0$. For any $\mathbf{x} \in S$, we have:

$$\nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) = 0^T(\mathbf{x} - \bar{\mathbf{x}}) = 0 \geq 0.$$

By the definition of pseudoconvexity, the condition $\nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ implies:

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}).$$

Since $\mathbf{x}$ was arbitrary in S , we have $f(\mathbf{x}) \geq f(\bar{\mathbf{x}})$ for all $\mathbf{x} \in S$, which means $\bar{\mathbf{x}}$ is a global minimizer. ✓

Intuition: The pseudoconvexity condition says “the gradient never lies.”

When the gradient is zero, it's telling you that you're at a minimum in every direction, so you've found the global minimum.

Proof Space: Convex implies Pseudoconvex

Theorem

If f is differentiable and convex on an open convex set S , then f is pseudoconvex on S .

Proof (Complete):

Assume f is convex on S . By the first-order characterization of convex functions (Lecture 8):

$$f(\mathbf{x}_2) \geq f(\mathbf{x}_1) + \nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) \quad \text{for all } \mathbf{x}_1, \mathbf{x}_2 \in S.$$

Now suppose $\nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) \geq 0$. Then:

$$\begin{aligned} f(\mathbf{x}_2) &\geq f(\mathbf{x}_1) + \nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) \\ &\geq f(\mathbf{x}_1) + 0 \\ &= f(\mathbf{x}_1). \end{aligned}$$

Thus, whenever the gradient condition holds, we have $f(\mathbf{x}_2) \geq f(\mathbf{x}_1)$, which is exactly the definition of pseudoconvexity. ✓

Key Insight: The first-order condition of convexity directly implies pseudoconvexity, because the “slope” is always being underestimated.

Proof Space: Pseudoconvex implies Strictly Quasiconvex

Theorem

If f is pseudoconvex on an open convex set S , then f is strictly quasiconvex on S .

Proof Outline:

Take $\mathbf{x}_1, \mathbf{x}_2 \in S$ with $f(\mathbf{x}_1) \neq f(\mathbf{x}_2)$. WLOG assume $f(\mathbf{x}_2) < f(\mathbf{x}_1)$.

By the contrapositive of pseudoconvexity:

$$f(\mathbf{x}_2) < f(\mathbf{x}_1) \quad \Rightarrow \quad \nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) < 0.$$

Now consider $\mathbf{x}_\lambda = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$ for $\lambda \in (0, 1)$. We have:

$$\mathbf{x}_\lambda - \mathbf{x}_1 = (1 - \lambda)(\mathbf{x}_2 - \mathbf{x}_1), \quad \text{so} \quad \nabla f(\mathbf{x}_1)^T (\mathbf{x}_\lambda - \mathbf{x}_1) = (1 - \lambda) \nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) < 0.$$

By the contrapositive of pseudoconvexity applied to the pair $\mathbf{x}_1, \mathbf{x}_\lambda$:

$$\nabla f(\mathbf{x}_1)^T (\mathbf{x}_\lambda - \mathbf{x}_1) < 0 \quad \Rightarrow \quad f(\mathbf{x}_\lambda) < f(\mathbf{x}_1).$$

Since $f(\mathbf{x}_1) = \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}$, we have $f(\mathbf{x}_\lambda) < \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}$ for all $\lambda \in (0, 1)$, which is the definition of strict quasiconvexity. ✓

Proof Space: First-Order Characterization of Quasiconvex

Theorem

f differentiable on open convex S is quasiconvex $\Leftrightarrow$

$$f(\mathbf{x}_2) \leq f(\mathbf{x}_1) \Rightarrow \nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) \leq 0 \text{ for all } \mathbf{x}_1, \mathbf{x}_2 \in S.$$

Proof Outline:

($\Rightarrow$) Assume f is quasiconvex. Suppose $f(\mathbf{x}_2) \leq f(\mathbf{x}_1)$ for some $\mathbf{x}_1, \mathbf{x}_2 \in S$.

For $\lambda \in (0, 1)$, by quasiconvexity:

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) \leq \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\} = f(\mathbf{x}_1).$$

The difference quotient is:

$$\frac{f(\mathbf{x}_1 + \lambda(\mathbf{x}_2 - \mathbf{x}_1)) - f(\mathbf{x}_1)}{\lambda} \leq 0.$$

Taking $\lambda \rightarrow 0^+$, the directional derivative $\nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) \leq 0$. ✓

($\Leftarrow$) Assume the gradient condition. Take $\mathbf{x}_1, \mathbf{x}_2 \in S$ with $f(\mathbf{x}_1) \geq f(\mathbf{x}_2)$ (the opposite case is symmetric). Then $f(\mathbf{x}_2) \leq f(\mathbf{x}_1)$ implies $\nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) \leq 0$.

For $\lambda \in (0, 1)$, let $\mathbf{x} = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$. By applying the condition twice and combining, we can show $f(\mathbf{x}) \leq f(\mathbf{x}_1)$, hence quasiconvexity. ✓

First-Order Characterization of Quasiconvex Functions

Pseudoconvex Functions

Key Properties of Pseudoconvex Functions

The Generalized Convexity Hierarchy

Pseudoconvex and Pseudoconcave: Pseudolinear Functions

Applications and Connections

Proof Space: Key Results

Strongly Convex Functions

Strongly Quasiconvex Functions

Other Generalizations of Convexity

End-of-Lecture Assessment

Strongly Convex Functions

Definition

Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$. The function f is **μ -strongly convex** with $\mu > 0$ if the function $g : S \rightarrow \mathbb{R}$, defined by

$$g(\mathbf{x}) \equiv f(\mathbf{x}) - \mu \|\mathbf{x}\|_2^2,$$

is convex. f is **μ -strongly concave** if $-f$ is μ -strongly convex.

Intuition: A strongly convex function is strictly convex with a quadratic lower bound. The parameter μ quantifies how quickly the function grows away from its minimizer.

Theorem (Hessian Characterization)

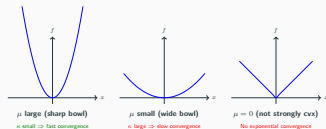
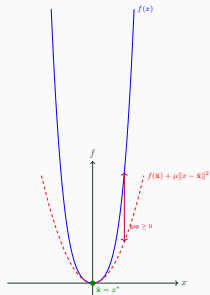
Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be twice differentiable on S . Then f is μ -strongly convex on S if and only if

$$\nabla^2 f(\mathbf{x}) \succeq 2\mu I_n \quad \text{for all } \mathbf{x} \in S.$$

Equivalently, all eigenvalues of $\nabla^2 f(\mathbf{x})$ are at least 2μ .

Visualizing Strong Convexity

A μ -strongly convex function sits above a quadratic bowl at every point:



Intuition: μ measures the minimum "curvature" of f . Larger μ means a tighter quadratic bowl, which makes gradient descent converge faster (fewer iterations to reach the bottom).

Why Strong Convexity Matters: Convergence Rates

Strong convexity is the *single most important structural property* for fast convergence of optimization algorithms:

The Convergence Rate Revolution

For gradient descent with step size η :

Function class	Convergence rate	To get ϵ -optimal
Convex	$O(1/t)$	$O(1/\epsilon)$ iterations
μ -Strongly convex	$O(e^{-\mu t/L})$	$O(\frac{L}{\mu} \ln \frac{1}{\epsilon})$ iterations

The **exponential vs. polynomial** difference is dramatic: μ -strong convexity means gradient descent converges *linearly* (in the optimization sense), not sublinearly.

The Condition Number $\kappa = L/\mu$

The ratio of the smoothness constant L (largest Hessian eigenvalue) to the strong convexity parameter μ (smallest Hessian eigenvalue) is the **condition number**. It determines:

- How many iterations gradient descent needs ($O(\kappa \ln 1/\epsilon)$)
- Whether accelerated methods (Nesterov) give significant speedup ($O(\sqrt{\kappa} \ln 1/\epsilon)$)
- How sensitive the solution is to numerical errors

Regularization Creates Strong Convexity

Adding ℓ_2 regularization: $\tilde{f}(\mathbf{x}) = f(\mathbf{x}) + \lambda \|\mathbf{x}\|^2$ makes *any* convex f into a λ -strongly convex function. This is one reason **ridge regression** and **weight decay** are so widely used — they guarantee fast convergence.

Examples of Strongly Convex Functions

Example (a): Consider $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ where A is positive definite.

The Hessian is $\nabla^2 f(\mathbf{x}) = 2A$. We need $2A - 2\mu I \succeq 0$, i.e., $A \succeq \mu I$. Thus f is μ -strongly convex whenever $0 < \mu \leq \lambda_{\min}(A)$.

Example (b): Consider $f(x) = \exp(x)$ on $[a, b]$.

We have $f''(x) = \exp(x) \geq \exp(a)$ on $[a, b]$. Thus f is μ -strongly convex on $[a, b]$ for $0 < \mu \leq \frac{\exp(a)}{2}$.

Non-example: $f(x) = x \ln(x)$ on $(0, +\infty)$.

We have $f''(x) = 1/x \rightarrow 0$ as $x \rightarrow \infty$, so f is *not* μ -strongly convex for any $\mu > 0$ on the full domain. However, f is μ -strongly convex on any bounded interval $[a, b] \subset (0, \infty)$ with $\mu \leq \frac{1}{2b}$.

Exercise 8: Strong Convexity

Quick Check

1. Determine the largest $\mu > 0$ such that $f(\mathbf{x}) = 2x_1^2 + 3x_1x_2 + 4x_2^2$ is μ -strongly convex.
2. Is $f(x) = x^4$ strongly convex on $\mathbb{R}$? What about on $[-1, 1]$?
3. True or False: Every μ -strongly convex function is strictly convex.

Hint: For (1), compute the Hessian eigenvalues. For (2), check the second derivative at $x = 0$.

Solution: Strong Convexity

Answers

(1) $\nabla^2 f = \begin{pmatrix} 4 & 3 \\ 3 & 8 \end{pmatrix}$. Eigenvalues: $\lambda = 6 \pm \sqrt{13}$. So

$\lambda_{\min} = 6 - \sqrt{13} \approx 2.39$. For μ -strong convexity, we need

$\lambda_{\min} - 2\mu \geq 0$, giving $\mu_{\max} = \frac{6 - \sqrt{13}}{2} \approx 1.20$.

(2) $f(x) = x^4$, $f''(x) = 12x^2$. At $x = 0$: $f''(0) = 0$. So

$f''(x) - 2\mu \geq 0$ fails at $x = 0$ for any $\mu > 0$. NOT strongly convex on $\mathbb{R}$.

On $[-1, 1]$: $\min_{x \in [-1, 1]} f''(x) = f''(0) = 0$. Still not strongly convex.
(Need $f''(x) > 0$ everywhere.)

(3) True. If f is μ -strongly convex, then $g(\mathbf{x}) = f(\mathbf{x}) - \mu\|\mathbf{x}\|^2$ is convex. So for $\lambda \in (0, 1)$:

$f(\lambda \mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2) \leq \lambda f(\mathbf{x}_1) + (1 - \lambda)f(\mathbf{x}_2) - \mu\lambda(1 - \lambda)\|\mathbf{x}_1 - \mathbf{x}_2\|^2 < \lambda f(\mathbf{x}_1) + (1 - \lambda)f(\mathbf{x}_2)$ when $\mathbf{x}_1 \neq \mathbf{x}_2$.

First-Order Characterization of Quasiconvex Functions

Pseudoconvex Functions

Key Properties of Pseudoconvex Functions

The Generalized Convexity Hierarchy

Pseudoconvex and Pseudoconcave: Pseudolinear Functions

Applications and Connections

Proof Space: Key Results

Strongly Convex Functions

Strongly Quasiconvex Functions

Other Generalizations of Convexity

End-of-Lecture Assessment

Strongly Quasiconvex Functions

Definition

Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$. The function f is **strongly quasiconvex** if for each $\mathbf{x}_1, \mathbf{x}_2 \in S$ with $\mathbf{x}_1 \neq \mathbf{x}_2$ and each $\lambda \in (0, 1)$:

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) < \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}.$$

f is **strongly quasiconcave** if $-f$ is strongly quasiconvex.

Key Implications:

- Strictly Convex $\Rightarrow$ Strongly Quasiconvex
- Strongly Quasiconvex $\Rightarrow$ Strictly Quasiconvex
- Strongly Quasiconvex $\Rightarrow$ Quasiconvex

Theorem (Unique Global Optimum)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be strongly quasiconvex and $S \subseteq \mathbb{R}^n$ be a nonempty convex set. If $\mathbf{x}^*$ is a local optimal solution to $\min_{\mathbf{x} \in S} f(\mathbf{x})$, then $\mathbf{x}^*$ is the *unique* global optimal solution.

Practical Application: Facility Location

In minimax facility location, $f(\mathbf{x}) = \max_i \|\mathbf{x} - \mathbf{a}_i\|$ (minimize the worst-case distance to demand points). This function is strongly quasiconvex on $\mathbb{R}^n$: for any two distinct points, the midpoint has a strictly smaller maximum distance. The theorem guarantees the optimal facility location is unique — any local search algorithm will find it.

Roadmap

First-Order Characterization of Quasiconvex Functions

Pseudoconvex Functions

Key Properties of Pseudoconvex Functions

The Generalized Convexity Hierarchy

Pseudoconvex and Pseudoconcave: Pseudolinear Functions

Applications and Connections

Proof Space: Key Results

Strongly Convex Functions

Strongly Quasiconvex Functions

Other Generalizations of Convexity

End-of-Lecture Assessment

Other Generalizations of Convexity

There are several other generalizations of convexity, some more useful than others:

- **Log-convex Functions:** Functions where $\ln f$ is convex. See Section 3.5 of Boyd and Vandenberghe.
- **Generalized Inequalities:** Convexity with respect to nonlinear cones. See Section 3.6 of Boyd and Vandenberghe.
- **L -convexity and M -convexity:** Discrete versions of convexity. See the Springer Encyclopedia article on Discrete Convex Analysis.
- **Invexity:** A generalization where the first-order condition characterizes global minima. See Wikipedia article on Invex functions.

Note: We will not focus much on these generalizations in this course. The key relationships covered in this lecture (convex $\subset$ pseudoconvex $\subset$ strictly quasiconvex $\subset$ quasiconvex, plus strong convexity and strong quasiconvexity) are the most important for algorithm design and convergence analysis.

Roadmap

First-Order Characterization of Quasiconvex Functions

Pseudoconvex Functions

Key Properties of Pseudoconvex Functions

The Generalized Convexity Hierarchy

Pseudoconvex and Pseudoconcave: Pseudolinear Functions

Applications and Connections

Proof Space: Key Results

Strongly Convex Functions

Strongly Quasiconvex Functions

Other Generalizations of Convexity

End-of-Lecture Assessment

Exercise 7: End-of-Lecture Assessment

Assessment: True/False and Conceptual

1. **True or False:** If f is convex and differentiable, then $\nabla f(\bar{\mathbf{x}}) = 0$ implies $\bar{\mathbf{x}}$ is a global minimizer.
2. **True or False:** Every quasiconvex function is pseudoconvex.
3. **True or False:** The sum of two pseudoconvex functions is pseudoconvex.
4. **Short Answer:** Explain why a fractional program $\min \frac{p(\mathbf{x})}{q(\mathbf{x})}$ with affine p and positive affine q has the property that every local minimum is a global minimum.
5. **True or False:** If f is pseudoconvex and $\mathbf{x}^*$ is a local minimizer, then $\mathbf{x}^*$ is a global minimizer.
6. **True or False:** Every strongly convex function has a unique global minimizer.

Solutions: End-of-Lecture Assessment

Answers

(1) **True.** If f is convex and $\nabla f(\bar{\mathbf{x}}) = 0$, then by convexity's first-order characterization:

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) = f(\bar{\mathbf{x}}) + 0 = f(\bar{\mathbf{x}}).$$

So $f(\mathbf{x}) \geq f(\bar{\mathbf{x}})$ for all $\mathbf{x}$, making $\bar{\mathbf{x}}$ a global minimizer. ✓

(2) **False.** Quasiconvex functions are more general than pseudoconvex. Example: $f(x) = x^3$ is quasiconvex but not pseudoconvex (at $x = 0$, gradient is zero but function decreases in negative direction).

(3) **False.** Unlike convex functions, pseudoconvex functions are not closed under addition. Counterexample can be constructed with carefully chosen functions. This is a key difference from convexity.

(4) The function $f(\mathbf{x}) = \frac{p(\mathbf{x})}{q(\mathbf{x})}$ is pseudolinear when p and q are affine and $q > 0$. Pseudolinear functions are both pseudoconvex and pseudoconcave. By the critical point property (Theorem 3), if $\nabla f = 0$, then $\mathbf{x}$ is a global min (and also a global max, but since f is pseudolinear, it's actually constant nearby). For all functions in the fractional programming class, local minima correspond to points where the gradient is zero (in the interior) or at the boundary. The pseudolinearity ensures these are global.

(5) **True.** This is Theorem 4 from the lecture. If $\bar{\mathbf{x}}$ is a local minimizer on convex S , then $\nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ for all $\mathbf{x} \in S$ (first-order optimality). By pseudoconvexity, this implies $f(\mathbf{x}) \geq f(\bar{\mathbf{x}})$ for all $\mathbf{x} \in S$. ✓

(6) **True.** Strong convexity implies strict convexity, which guarantees that if a minimizer exists, it is unique. Moreover, on $\mathbb{R}^n$, strong convexity ensures the objective grows without bound (coercivity), guaranteeing existence.

Key Takeaways

Main Results

- **First-Order Characterization of Quasiconvex:**
 $f(\mathbf{x}_2) \leq f(\mathbf{x}_1) \Rightarrow \nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) \leq 0$
- **Pseudoconvexity Definition:**
 $\nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) \geq 0 \Rightarrow f(\mathbf{x}_2) \geq f(\mathbf{x}_1)$ (gradient never lies)
- **Critical Point Property:** If f is pseudoconvex and $\nabla f(\bar{\mathbf{x}}) = 0$, then $\bar{\mathbf{x}}$ is a global minimizer
- **Local $\Rightarrow$ Global:** For pseudoconvex f on convex domain, every local minimum is global
- **Generalized Hierarchy:** Convex $\subset$ Pseudoconvex $\subset$ Strictly Quasiconvex $\subset$ Quasiconvex
- **Pseudolinear Functions:** Ratios with affine numerator/denominator are pseudolinear (fractional programming)
- **Strong Convexity:** $f(\mathbf{x}) - \mu\|\mathbf{x}\|^2$ is convex; guarantees unique minimizer and fast convergence
- **Strong Quasiconvexity:** $f(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2) < \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}$; unique global optimum

Connections to Optimization

- Pseudoconvex $\Rightarrow$ gradient descent converges to global minimum
- Fractional programming is pseudolinear $\Rightarrow$ specialized algorithms available
- Understanding the hierarchy guides algorithm selection
- Pseudoconvexity is harder to verify than convexity but broader in scope

Next Time: Lecture 10

Lecture 10 Preview: Optimality Conditions and Line Search Methods

- Descent directions and first-order necessary conditions
- Second-order necessary and sufficient conditions for optimality
- Introduction to line search methods
- Derivative-free methods: golden section, Fibonacci search
- Derivative-based methods: steepest descent (1D), Newton's method (1D)

Big Picture: Today we explored the landscape of generalized convexity classes and learned that pseudoconvex functions combine many advantages of convex functions (global minima are stationary points) with broader applicability (fractional programming, certain nonconvex problems). Next, we'll see how this theory translates into practical algorithms for finding these global minima efficiently.

Questions?

Office hours: See syllabus. Feel free to ask about pseudoconvex functions, the generalized convexity hierarchy, or connections to algorithms!